

The optional infinitive stage in Japanese*

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1. Introduction

In adult Japanese, subjects can be marked with *-ga* (nominative case) or *-no* (genitive case). This case alternation is possible only in embedded contexts headed by a nominal expression, such as a nominal complement clause or a relative clause, as shown in (1); genitive subjects never appear in the matrix clause, as shown in (2).

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|--|---|
| (1) [John-ga/no yonda] hon
John-NOM/GEN read-PAST book
'the book that John read' | (2) John-ga/*no hon-o yonda.
John-NOM/GEN book-ACC read
'John read the book.' |
|--|---|

In child Japanese, however, genitive subjects have been observed to show up in the matrix clause, as exemplified in (3) (Suzuki 2001, 2007, Sawada et al. 2010, Sawada and Murasugi 2011).

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|---|------------------------|
| (3) Taisyoo-kun-no tukut-ta. (Tai, 1;10)
Taishoo-kun-GEN make-PAST
'Mr. Taishoo made (this).' | (Sawada et al. 2010:3) |
|---|------------------------|

In this study, we explore the reason why the genitive subject can occur in the matrix clause in child Japanese. We argue that the matrix genitive subject in child Japanese is an instance

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of the Optional Infinitive Stage (OIS) of language development (Wexler 1994, 1998). We propose a modification to the feature combination in Wexler's OIS work in order to adopt it for a language without phi-feature agreement.

2. Genitive subjects in adult Japanese

Miyagawa (2011) argues that the genitive subject remains within vP (see also Watanabe 1996 and Ochi 2017). One piece of evidence is found in example (4), in which the nominative subject, but not the genitive subject, can precede vP adverbs.

- (4) [Kodomo-tati-ga/*no minnade ikioiyoku kake-nobot-ta] kaidan
 child-PL-NOM/GEN together vigorously run-climb.up-PAST stairway
 'the stairway which those children ran up together vigorously' (Harada 1971:30)

Miyagawa (2011) argues that the nominative case feature originates at C and is inherited by T, giving T the ability to assign the nominative case along with the EPP feature; thus, the subject is raised to Spec-T and can precede vP adverbs. In contrast, the genitive subject appears in TP without CP, and the lack of C renders T defective, making it unable to assign the nominative case to the subject or bear the EPP feature. The absence of the EPP forces the genitive subject to remain in the vP-internal position, with the consequence that the genitive subject must follow the vP adverb, as we observed in (4). Due to the reduced size of the clause (i.e., the absence of the phase CP layer), the nominal head above TP can license the genitive case on the subject in Spec-v. Miyagawa's analysis is sketched in (5), where "D" represents a nominal head taking a clause as its complement.

- (5) a. [DP [D' [CP [C' [TP subject-NOM_i [T' [vP *t_i* ...] T]] C]] D]]
 b. [DP [D' [TP [T' [vP subject-GEN ...] T]] D]]

Further evidence for the vP-internal genitive subject analysis is based on cases like (6a). The nominative subject can co-occur with the accusative object, while the genitive subject cannot (Harada 1971). The ungrammaticality of (6a) is mitigated by relativizing the accusative object *LGB-o* 'LGB-ACC', as demonstrated in (6b) (Harada 1971).

- (6) a. [Kinoo John-ga/*no LGB-o katta] mise
 yesterday John-NOM/GEN LGB-ACC bought shop
 'the shop where John bought LGB yesterday'
 b. [Kinoo John-ga/no katta] LGB
 yesterday John-NOM/GEN bought LGB
 'LGB that John bought yesterday'

This restriction on the co-occurrence of the genitive subject with the accusative object has come to be called the transitivity restriction (Watanabe 1996). Alexiadou and

Anagnostopoulou (2007) propose that the transitivity restriction can be accommodated by their subject-in-situ generalization (SSG) in (7).

(7) *The subject-in-situ generalization (SSG)*

By Spell-Out, vP can contain only one argument with a structural Case feature.

(Alexiadou and Anagnostopoulou 2007:31, fn. 1 omitted)

Example (6a) is ungrammatical when the subject is marked with the genitive case because the genitive subject and the accusative object both stay in vP in violation of the SSG.

3. The optional infinitive stage in child languages

3.1. The Agreement and Tense Omission Model: Schütze and Wexler (1996)

Children aged around two optionally produce an infinitival form of a verb in the main clause; they can also use the expected finite form of a verb during the same period. This use of the infinitive in the main clause is referred to as the OIS (Wexler 1994, 1998).

In this stage, English-speaking children often produce agreement-less or tenseless clauses. Schütze and Wexler (1996) propose the Agreement (AGR) and Tense (TNS) Omission Model (ATOM) to explain the OIS and suggest that children may freely omit features on the AGR head and/or the TNS head, yielding four possible feature combinations, with each feature having the “+/-” bifurcation: [+TNS, +AGR], [+TNS, –AGR], [–TNS, +AGR], [–TNS, –AGR]. Based on the ATOM, the nominative case is assigned to a subject in the case of [+TNS, +AGR] or [–TNS, +AGR], while the genitive case is given to a subject in the case of [–TNS, –AGR].

3.2. A possible extension to child Japanese

Sawada and Murasugi (2011) argue that the presence of the matrix genitive subject in child Japanese is analyzed as an instance of the OIS, an idea that we endorse. However, they do not offer an analysis of why the genitive subject may occur in the matrix clause in the OIS. Given that Japanese does not have phi-feature agreement within CP (e.g., Saito 2007, Miyagawa 2010, 2017), we need to modify the original ATOM schema, which contains [AGR], so that it is applicable to Japanese.

One avenue for explaining the OIS stage in Japanese is to appeal to the D feature on T responsible for EPP movement (Chomsky 1995). Let us suppose that, in Japanese, the D feature originates at C, and is inherited by T, which triggers EPP movement of the subject to Spec-T (Miyagawa 2001). We argue that the feature combination for the ATOM in Japanese is D and TNS in place of AGR and TNS in languages with phi-feature agreement such as English. The four possible feature combinations in Japanese are thus as follows:

- (8)
- a. [+TNS, +D] → nominative subject
 - b. [+TNS, -D] → genitive case (OIS)
 - c. [-TNS, +D] → genitive case (OIS)
 - d. [-TNS, -D] → genitive case (OIS)

[+TNS, +D] is the same as adult Japanese, in which the subject is assigned the nominative case. All others are instantiations of the OIS, with the genitive case on the subject. In the following, we account for the genitive case marking on the matrix subject in child Japanese, drawing on the feature specifications given in (8).

4. Proposal: The OIS and case marking on subjects in child Japanese

According to Sawada et al. (2010), 4% of the matrix subjects are non-nominative during the relevant acquisition period, and of these, 58% were genitive (others were dative or accusative). Building on Sawada et al.'s (2010) and Dansako's (2020) surveys, we also compiled 5 Japanese-speaking children's data (*Aki*, *Ryo*, *Tai*, *Jun*, *Sumihare*) by using CHILDES (MacWhinney 2000) and one child, *Child A*, by referring to Suzuki's (2001, 2007) papers. The following table shows, for all the children we examined, the number of matrix genitive subjects based on the four possible feature combinations in (8), with the position of adverbs indicating whether the subject has moved out of the vP or not.

(9) Possible feature combinations and the number of matrix genitive subjects

	[+TNS]	[-TNS]
[+D]	8	2
[-D]	47	32
The number of genitive subjects		89

Based on our study, there are three feature combinations of the OIS. The first case is [+TNS, -D], in which the verb is inflected with the past tense morphology, *-ta*, or present tense morpheme, as shown in (10a).

- (10)
- a. Kokoni Tai-tyan-no wakat-ta. (Tai, 2;01)
here Tai-tyan-GEN understand-PERF
Int: 'Tai-tyan has got (it) here.'
 - b. Ima A-tyan-no tukut-ten no. (Child A, 2;04)
now A-tyan-GEN make-PROG SFP
'Now, A-tyan is making (this).'
- (Suzuki 2001:52)

Examples of this sort indicate that the subject stays in vP because of the absence of the D feature on T. Thus, the genitive subject is preceded by a locative PP, as in (10a), or a temporal adverb (*ima* 'now'), as in (10b).

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The second case is [–TNS, +D], in which [–TNS] points to an atelic and nondynamic meaning, that is, a stative interpretation (Miyagawa 2011). As shown in (11), the genitive subject unexpectedly has moved out of a vP, thus the [+D] designation, which avoids a SSG violation. We will return to the discussion of this case later.

- (11) Henry-no anoo suna-o hakonde-it-tyau. (Tai, 2;07)
Henry-GEN well sand-ACC carry-go-PERF
'Henry has carried sand.'
(Suzuki 2001:53)

The third case is [–TNS, –D], in which verbs are also categorized as having a stative aspect ([–TNS]), and the genitive subject follows a locative expression because of the absence of the D feature on T, which forces the subject to stay in Spec-v, as shown in (12).

- (12) a. Kore, masukuman-no i-ku. (Ryo, 2;11)
this mask.man-GEN go-PRES
'Here, Mask-man goes.'
(Sawada et al. 2010:3)
- b. Datte koko hikooki-no toor-u mon. (Tai, 2;10)
because here airplane-GEN go.through-PRES SFP
Lit: 'That's because an airplane flies over there.'

Recall that in adult Japanese, the genitive subject is limited to a nominal complement clause or a relative clause. However, as we have seen, in child Japanese the genitive subject can occur in the matrix clause. Not only that, there is an indication that the matrix genitive subject occurs with the CP (instead of just the TP), a clear contrast to adult Japanese. The first piece of evidence that it occurs in CP comes from the fact that it can occur in a question construction, as shown in (13), in which *titi* 'father' is marked with the genitive case, and the clause is headed by the sentence-final question particle *-no*.

- (13) Titi-no patin sur-u no? (Child A, 2;05)
father-GEN onomatopoeia do-PRES Q
'Does my father lock the bicycle (security) belt?'
(Suzuki 2001:54)

Assuming that *yes/no* questions are CPs, examples of this kind show that the genitive subject appears in CP in child Japanese. The second piece of evidence is based on the presence of sentence final particles, which we saw in (10b) and (12b) above. Sentence final particles are assumed to be merged in the CP zone (or higher) in Japanese (see Saito 2015 and Miyagawa 2022).

An additional property of the OIS genitive subject that distinguishes it from adult Japanese is the ability to avoid the SSG. As we saw earlier in (11), the genitive subject may occur with an accusative object, which is not possible in adult Japanese due to the SSG. This is made possible by the occurrence of C, where the D feature originates and is

inherited by T. This is something that is not observed in adult Japanese; it is attested only in the OIS stage.

Further evidence that the matrix genitive subject is realized in CP is shown by example (13). The genitive subject occurs away from the verb, with the onomatopoeic expression *patin* intervening between the two. This example and example (11) above suggest that C occurs to make it possible for the D feature to be transmitted to T, which results in the genitive subject undergoing movement to Spec-T.

Another piece of evidence that C occurs with the matrix genitive subject is the possibility of scrambling the object across the genitive subject. As shown in the examples in (14), the object can scramble over the genitive subject in child Japanese.

- (14) a. Kore, A-tyan-no tukut-ta no. (Child A, 2;02)
 this A-tyan-GEN make-PAST SFP
 ‘This, A-tyan made.’ (Suzuki 2001:52)
- b. Kore, Taisyoo-kun-no tukut-ta. (Tai, 1;10)
 this Taisyoo-kun-GEN make-PAST
 ‘This, Mr. Taisyoo made.’

In adult Japanese, however, scrambling of the object is blocked in the presence of a genitive subject, as shown in (15) (Dubinsky 1993). (Following Miyagawa (2013, 2017), we modified the original example provided by Dubinsky 1993 to avoid a violation of the SSG.)

- (15) [Geki-de; musuko-ga/*no *t_i* odotta] koto
 play-in son-NOM/GEN danced fact
 ‘the fact that my son danced in the play’

Miyagawa (2010) suggests that scrambling is available only when T is selected by C, as T can inherit formal features, including the EPP (D) feature, from C, which trigger movement, including scrambling (Miyagawa 2001). Miyagawa (2013, 2017) attributes the ungrammaticality of (15) to the absence of C; in adult Japanese, a sentence is projected only up to TP when the genitive subject occurs, and nothing licenses scrambling. In contrast, scrambling of the object over the genitive subject is possible in child Japanese, as we observed. This striking fact is accounted for by the presence of C: in the examples in (14), the D feature, which drives movement, is inherited by T from C, which makes it possible for scrambling across the genitive subject to take place.

It should be noted that children also produce the nominative subject during the same period when they use the genitive subject in the matrix clause. See example (16).

- (16) Usagi-ga kasiko siteru wa. (Jun, 2;11)
 rabbit-NOM wisely do.PROG SFP
 Lit: ‘The rabbit is wisely doing (something).’

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Examples like this reflect the [+TNS, +D] pattern; this is the same as in adult Japanese. By occurring within the CP, T inherits both the ability to assign the nominative case to the subject and the D feature that activates movement of the subject to Spec-T. Notice that *kasiko* ‘wisely’ (the adult form: *kasikoku*) in (16) is a vP adverb; the nominative subject *usagi-ga* ‘the rabbit’ in (16) undergoes raising to Spec-T from within the verb phrase, as it precedes this adverb. This adverb cannot occur between the genitive subject and the verb in adult Japanese, as shown in (17).

- (17) [Kinoo John-ga/*no kasikoku tukatta] konpyuutaa.
yesterday John-NOM/GEN wisely used computer
‘the computer that John wisely used yesterday’

Therefore, the nominative subject is realized only when both the TNS and the D features are specified as “+”.

There are complications when it comes to the interpretation of the D feature and the TNS feature. As indicated in the table in (9), [+TNS, +D], which is for nominative subject in both adult and child Japanese, also shows up unexpectedly in 8 examples with the genitive subject in child Japanese. The following are two such examples:

- (18) a. Neko-tyan-no yooguruto tabe-te-ta. (Tai, 2;00)
cat-tyan-GEN yogurt eat-PROG-PAST
‘The cat has been eating yogurt.’
b. Kore, Taisyoo-kun-no tukut-ta. (= (14b)) (Tai, 1;10)
this Taisyoo-kun-GEN make-PAST
‘This, Mr. Taisyoo made.’

These examples indicate that the D feature occurs on T; thus, it is fully tensed with the CP. In (18a), the subject occurs with an accusative object; the subject must have moved to Spec-T to avoid the SSG, a movement made possible by the D feature on T. In (18b), the object, *kore* ‘this’, has scrambled across the subject, which avoids the SSG, made possible by the occurrence of the D feature on T. These are fully tensed clauses with a D feature on T, which is typically associated with the nominative subject, yet, in these cases, the subject occurs with the genitive case.

We do not know why the genitive subject occurs in these examples. One possibility, though admittedly not very interesting, is that these are simply mistaken uses of the genitive case. The reason why we raise this possibility is that these examples were observed at the very early stages only for two children, *Tai* and *Child A*: 7 of 8 examples were produced at 1;10-2;01 (*Tai*) and at 2;02-2;04 (*Child A*). Only one example that *Tai* produced was observed relatively later (2;07). It is possible that these two children may not have known exactly what feature combination governs the occurrence of the genitive subject at that

stage. At this point, this possibility remains a speculation, which we hope to investigate more fully in the future.

5. Discussion

Our analysis is based on the assumption that the D feature is necessary for movement of the subject to Spec-T in both child and adult Japanese. As suggested by Chomsky (1995), the D feature occurs in agreement languages to trigger EPP movement. Our analysis, coupled with Chomsky's (1995) analysis, naturally leads to the suggestion that the D feature is the most fundamental agreement feature across all languages. Example (19) from Harris and Wexler (1996) is highly suggestive of this.

(19) Fraser not see him. (Eve, 2;00)

(Harris and Wexler 1996:34)

Neither tense nor agreement occurs in this example, yet the subject moves across negation (presumably) to Spec-T. This suggests that the EPP is activated in the presence of [+D] on T, independent of phi-feature agreement. In fact, the ATOM analyzes an English sentence like (19) as [+TNS, -AGR] or [-TNS, +AGR]. Since the third-person singular -s in English demands both [+AGR] and [+TNS], -s does not appear. If at least one of [+AGR]/[+TNS] precedes negation, example (19) will result. Our results here point to a somewhat different way to look at this phenomenon.

Another implication of our analysis is the licensing mechanism for the genitive subject in child Japanese. Recall that the OIS genitive case marking on the subject occurs without any nominal head, unlike the way the genitive case is assigned in adult Japanese by a D head. We suggest that the genitive case shows up as the default case in the absence of a fully specified set of features, similar to the accusative case in English.¹

Lastly, the fact that the OIS occurs in a language without agreement implies that it is a universal phenomenon, rather than being restricted to agreement languages like English. A question that arises is whether null-subject licensing Romance languages like Italian or Spanish, which have been argued to be without the OIS stage (Wexler 1998), in fact have it, but in some other form than the infinitive, as we saw for Japanese.

We should point out one empirical issue that needs a deeper analysis. The rate of genitive subjects in child Japanese is much smaller than the rate of non-finite instead of finite verbs in OIS languages. So far, we have not reached a satisfying explanation for this difference. Since the ATOM follows from a deeper theory, the Unique Checking Constraint (Wexler 1998), one avenue of explanation might be to try to understand the use of the D-

¹ See also Moritake and Maeda (2024), who reach a similar conclusion on independent grounds; to be more specific, the genitive case marking in a cleft construction in adult Japanese may be governed by the default strategy.

feature in that theory compared to this current work. One further hint toward an explanation may be that the overall percentage of genitive subjects in the relevant embedded clauses in adult Japanese is only about 10% (Nambu and Nakatani 2023). Other possible causes should be explored, which we leave for future investigation.

6. Conclusion

In this study, we addressed a fundamental difference between child Japanese and adult Japanese; the former, but not the latter, allows the genitive subject in the matrix clause. We suggested a possible way to answer this question by associating the OIS stage of language development with the presence of the matrix genitive subject in child Japanese. Our proposal is that four possible feature combinations using the [TNS] and [D] features make it possible for the presence of the genitive subject in the matrix clause in child Japanese.

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